

MTH166

Lecture-12

**Solution of Higher Order Homogeneous
LDE with Constant Coefficients-II**

Topic:

Solution of Higher order Homogeneous LDE with Constant coefficients

Learning Outcomes:

1. Formulation of 3rd order and 4th order homogeneous LDE when roots are given.

Formulation of LDE: $ay''' + by'' + cy' + dy = 0$ when Roots are given:

Let the three given roots be: m_1, m_2 and m_3 .

Then required 3rd order homogeneous LDE is:

$$y''' - (\text{sum of roots taken one at a time}) y'' + (\text{sum of roots taken one at a time}) y' - (\text{Product of roots}) y = 0$$

$$\text{i.e. } y''' - (m_1 + m_2 + m_3)y'' + (m_1m_2 + m_2m_3 + m_3m_1)y' - (m_1m_2m_3)y = 0$$

or

Then required 3rd order homogeneous LDE is:

$$(D - m_1)(D - m_2)(D - m_3)y = 0 \text{ where } D \equiv \frac{d}{dx}$$

Formulation of LDE: $ay^{iv} + by'''' + cy'' + dy' + ey = 0$ when Roots are given:

Let the four given roots be: m_1, m_2, m_3 and m_4 .

Then required 4th order homogeneous LDE is:

$$(D - m_1)(D - m_2)(D - m_3)(D - m_4)y = 0 \text{ where } D \equiv \frac{d}{dx}$$

Find a homogeneous LDE with constant coefficients of lowest order which has the following particular solution:

Q 1. $5 + e^x + 2e^{3x}$

Sol. Here: $5 + e^x + 2e^{3x} = 5e^{0x} + e^{1x} + 2e^{3x}$

So, $m_1 = 0, m_2 = 1, m_3 = 3$

Then required 3rd order homogeneous LDE is:

$$y''' - (m_1 + m_2 + m_3)y'' + (m_1m_2 + m_2m_3 + m_3m_1)y' - (m_1m_2m_3)y = 0$$

$$\Rightarrow y''' - (0 + 1 + 3)y'' + (0 + 3 + 0)y' - (0)y = 0$$

$$\Rightarrow y''' - 4y'' + 3y' = 0 \quad \textbf{Answer.}$$

Q 2. $xe^{-x} + e^{2x}$

Sol. Here: $xe^{-x} + e^{2x} = (0 + 1x)e^{-x} + e^{2x} \quad [(c_1 + c_2x)e^{m_1x} + c_3e^{m_2x}]$

So, $m_1 = -1, m_2 = -1, m_3 = 2$

Then required 3rd order homogeneous LDE is:

$$y''' - (m_1 + m_2 + m_3)y'' + (m_1m_2 + m_2m_3 + m_3m_1)y' - (m_1m_2m_3)y = 0$$

$$\Rightarrow y''' - (-1 - 1 + 2)y'' + (1 - 2 - 2)y' - (2)y = 0$$

$$\Rightarrow y''' - 3y' - 2y = 0 \quad \textbf{Answer.}$$

Q 3. $e^{-x} + \cos 5x + 3 \sin 5x$

Sol. Here: $e^{-x} + \cos 5x + 3 \sin 5x = c_1 e^{m_1 x} + e^{\alpha x} [c_2 \cos \beta x + c_3 \sin \beta x]$

So, $m_1 = -1, m_2 = 0 + 5i, m_3 = 0 - 5i$

Then required 3rd order homogeneous LDE is:

$$y''' - (m_1 + m_2 + m_3)y'' + (m_1 m_2 + m_2 m_3 + m_3 m_1)y' - (m_1 m_2 m_3)y = 0$$

$$\Rightarrow y''' - (-1 + 5i - 5i)y'' + (-5i + 25 + 5i)y' - (25i^2)y = 0$$

$$\Rightarrow y''' + y'' + 25y' + 25y = 0 \quad \textbf{Answer.}$$

Q 4. $1 + x + e^x - 3e^{3x}$

Sol. Here: $1 + x + e^x - 3e^{3x} = (1 + x)e^{0x} + e^x - 3e^{3x}$

So, $m_1 = 0, m_2 = 0, m_3 = 1, m_4 = 3$

Then required 4th order homogeneous LDE is:

$$(D - m_1)(D - m_2)(D - m_3)(D - m_4)y = 0 \quad \text{where } D \equiv \frac{d}{dx}$$

$$\Rightarrow (D - 0)(D - 0)(D - 1)(D - 3)y = 0$$

$$\Rightarrow D^2(D^2 - 4D + 3)y = 0$$

$$\Rightarrow (D^4 - 4D^3 + 3D^2)y = 0$$

$$\Rightarrow y^{iv} - 4y''' + 3y'' = 0 \quad \textbf{Answer.}$$

Polling Questions:

Q1. If e^x, e^{4x} are solutions of differential equation $y'' + a(x)y' + b(x)y = 0$, then the values of $a(x)$ and $b(x)$ are:

(A) $a(x) = -5, b(x) = 4$

(B) $a(x) = 5, b(x) = 4$

(C) $a(x) = -5, b(x) = -4$

(D) $a(x) = 5, b(x) = -4$

Q2. The intervals on which the differential equation $y' = 3\frac{y}{x}$ is normal are:

(A) $(-\infty, 0), (0, \infty)$

(B) $(-\infty, \infty)$

(C) $(-\infty, 1), (1, \infty)$

(D) None of these.

Q3. If $y = e^{at}$ is solution of $y'' - 5y' + 4y = 0$, then possible value of a is:

(A) $a = 2$

(B) $a = 3$

(C) $a = 4$

(D) $a = 5$

Q4. The general solution of $y'' - 9y = 0$ is:

(A) $Ae^{-3x} + Be^{3x}$

(B) $Ae^{3x} + Be^{3x}$

(C) $Ae^{-3x} + Be^{-3x}$

(D) None of these

Q5. The general solution of $y'' + 4y = 0$ is:

(A) $(A \cos 2x + B \sin 2x)$

(B) $Ae^{2x} + Be^{-2x}$

(C) $(A + Bx)e^{-2x}$

(D) None of these

Q6. The Wronskian of functions: $(1, \sin x, \cos x)$ is:

(A) 0

(B) 1

(C) -1

(D) None of these

Q7. The general solution of $y'' - 10y' + 25y = 0$ is:

(A) $Ae^{4x} + Be^{5x}$

(B) $Ae^{5x} + Be^{5x}$

(C) $Ae^{4x} + Be^{7x}$

(D) $Ae^{5x} + Bxe^{5x}$

Q15. The second order linear homogeneous differential equation with variable coefficients is

(a) $y'' + 9y' + y = \log(x^2 - 9)$

(b) $(1 + x^2)y'' + 2xy' + y = 0$

(c) $(1 + x^2)y'' + 2xy' + y = x^2$

(d) $(1 + x^2)y'' + 2yy' + 3y = 0$

Q20. The differential equation $yy'' + 2y' + y = \sin x$ is...

(a) linear

(b) homogeneous

(c) non-linear

(d) of order 3

Q21. The complementary function differential of the equation $(D^2 - 2D + 5)^2 y = 0$ is

(a) $e^x(A\cos 2x + B\sin 2x)$

(b) $e^x\{(A + Bx)\cos 2x + (C + Dx)\sin 2x\}$

(c) $e^x(A\cos 2x + B\sin 2x) + e^x(C\cos 2x + D\sin 2x)$

(d) None of these

